

SEMINAR 3 (Pilot). LIMITS OF SEQUENCES. COMPUTATIONS

You can use the following limits

$$\lim_{n \rightarrow \infty} a^n = 0, 0 < a < 1; \quad \lim_{n \rightarrow \infty} \frac{n^k}{a^n} = 0, a > 1, k \in \mathbb{N}; \quad \lim_{n \rightarrow \infty} \sqrt[n]{a} = 1; \quad \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1.$$

1. Find the following limits:

$$\begin{aligned} \text{(a)} \quad \lim_{n \rightarrow \infty} \frac{\sqrt{n^2 + n + 5}}{n + 1}; \quad & \text{(b)} \quad \lim_{n \rightarrow \infty} \frac{(-3)^n + 7^{n+1}}{(-3)^{n+2} + 7^n}; \quad & \text{(c)} \quad \lim_{n \rightarrow \infty} \frac{5n^2 + \sqrt{n^3 + 5n} - n + 3}{\sqrt{n^4 - 3n + 7} + \sqrt{n - 1}}; \\ & \text{(d)} \quad \lim_{n \rightarrow \infty} (n^2 - 2n + 5); \quad & \text{(e)} \quad \lim_{n \rightarrow \infty} \frac{5n^2 + n + 2}{1 + 2 + \dots + n}; \\ & \text{(f)} \quad \lim_{n \rightarrow \infty} \frac{n^3 + 3^n}{n + 3^{n+1}}; \quad & \text{(g)} \quad \lim_{n \rightarrow \infty} \sqrt[n]{3^n + n \cdot 2^n}. \end{aligned}$$

2. (HW) Find the following limits:

$$\begin{aligned} \text{(a)} \quad \lim_{n \rightarrow \infty} \frac{n^2 + 4n - 11}{3n^3 - 4n^2 + 5n - 2}; \quad & \text{(b)} \quad \lim_{n \rightarrow \infty} \frac{5n^4 - 6n + 2}{-n^2 + n - 1}; \quad & \text{(c)} \quad \lim_{n \rightarrow \infty} \frac{\sqrt{3n^4 + 2n^3 + 5}}{n^2 + 7}; \\ \text{(d)} \quad \lim_{n \rightarrow \infty} \frac{1 - n + 2n^2}{2 + 4 + \dots + 2n}; \quad & \text{(e)} \quad \lim_{n \rightarrow \infty} \frac{n^{10} - 1}{1 + n \cdot 1.1^n}; \quad & \text{(f)} \quad \lim_{n \rightarrow \infty} \left(\frac{-5n + 4n^2 - 4}{n - 5} - \frac{4n^2 - 3}{n + 4} \right). \end{aligned}$$

3. (HW) Find the following limits:

$$\text{(a)} \quad \lim_{n \rightarrow \infty} \frac{-5n^8 + n - 6}{\sqrt{6n^{16} + 7n - 6} + \sqrt{7n^8 - 3}}; \quad \text{(b)} \quad \lim_{n \rightarrow \infty} \sqrt[n]{\frac{n^2 + 4^n}{n + 5^n}}.$$

4. (HW) Find the following limits:

$$\text{(a)} \quad \lim_{n \rightarrow \infty} \frac{2 \cdot 5^n - 5 \cdot 4^n + 3}{4 \cdot 5^n - 3 \cdot 4^n + 2}; \quad \text{(b)} \quad \lim_{n \rightarrow \infty} \frac{2 \cdot 6^{-n} + 5 \cdot 5^{-n}}{4 \cdot 5^{-n} - 3 \cdot 6^{-n}}.$$

5. Find the following limits:

$$\text{(a)} \quad \lim_{n \rightarrow \infty} \left(\sqrt{n + 2025} - \sqrt{n} \right); \quad \text{(b)} \quad \text{(HW)} \quad \lim_{n \rightarrow \infty} \left(\sqrt{n^2 + 3n - 1} - \sqrt{n^2 - n} \right).$$

6. Compute $\lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} \right).$

7. (HW) Find the following limits:

$$\text{(a)} \quad \lim_{n \rightarrow \infty} \left(\frac{n}{n} - \frac{n-1}{n} + \frac{n-2}{n} - \dots + (-1)^{n-1} \frac{1}{n} \right); \quad \text{(b)} \quad \lim_{n \rightarrow \infty} \left(\sqrt[3]{3} \cdot \sqrt[9]{3} \cdot \sqrt[27]{3} \dots \sqrt[3^n]{3} \right).$$

8*. (HW) Compute the limit

$$\lim_{n \rightarrow \infty} \left(\frac{1^2}{n^3} + \frac{2^2}{n^3} + \dots + \frac{(n-1)^2}{n^3} \right).$$